

Indices

Worksheet 2 (Calculator)

1. W09/qp2/q1(a)

Solve

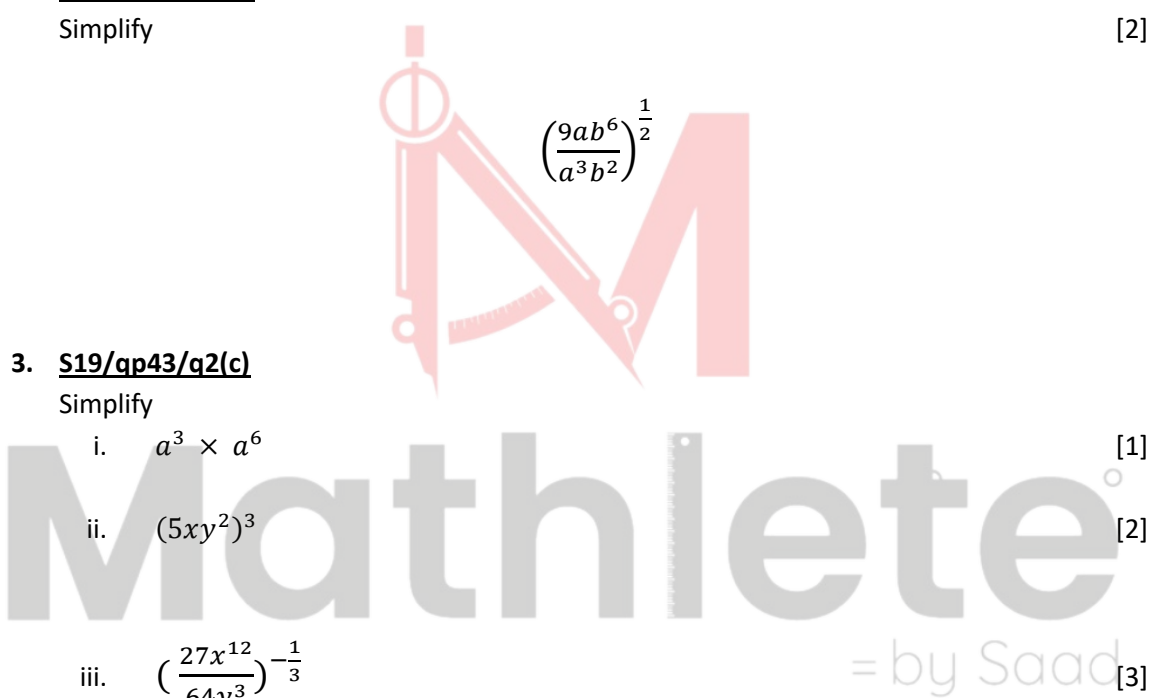
[1]

$$2^y = 8$$

2. S16/qp21/q2(b)

Simplify

[2]


$$\left(\frac{9ab^6}{a^3b^2}\right)^{\frac{1}{2}}$$

3. S19/qp43/q2(c)

Simplify

i. $a^3 \times a^6$

[1]

ii. $(5xy^2)^3$

[2]

iii. $\left(\frac{27x^{12}}{64y^3}\right)^{-\frac{1}{3}}$

[3]

4. S21/qp21/q3(d)

$$\frac{x^6}{x^m} = x^{-3}$$

Find m.

[1]

5. W22/qp42/q6(c and d) (0580)

- a. Write as a single fraction in its simplest form.

[3]

$$\frac{4a}{5} \div \frac{3ad}{10c}$$

- b. $5^n + 5^n + 5^n + 5^n + 5^n = 5^m$

Find an expression for m in terms of n.

[2]

6. W22/qp43/q2

Simplify

i. $p^3 \times p^{11}$

[1]

ii. $\frac{18m^6}{3m^2}$

[2]

iii. $\left(\frac{27x^9y^{27}}{64}\right)^{-\frac{1}{3}}$

[3]

7. S23/qp42/q9(a) (0580)

Simplify

i. $(3x^2y^4)^3$

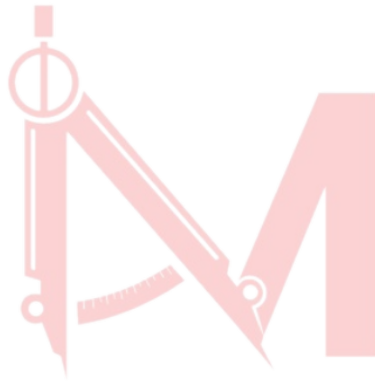
[2]

ii. $\left(\frac{16}{x^{16}y^8}\right)^{-\frac{3}{2}}$

[3]

Answer Key:

1. 3
2. $\frac{3b^2}{a}$
3. (i) a^9 (ii) $125x^3y^6$ (iii) $\frac{4y}{3x^4}$
4. 9
5. (c) $\frac{8c}{3d}$ (d) $n+1$
6. (a)(i) p^{14} (a)(ii) $6m^4$ (iii) $(\frac{4}{3x^3y^9})^{-1/3}$
7. (a)(i) $27x^6y^{12}$ (ii) $\frac{x^{24}y^{12}}{64}$



Mathlete
= by Saad